

Mainboard

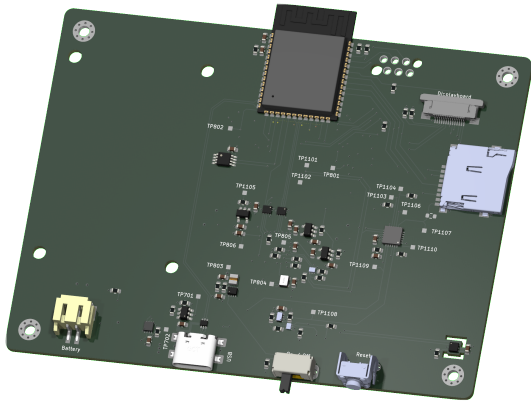
Variant: CHECKED

2026-02-23

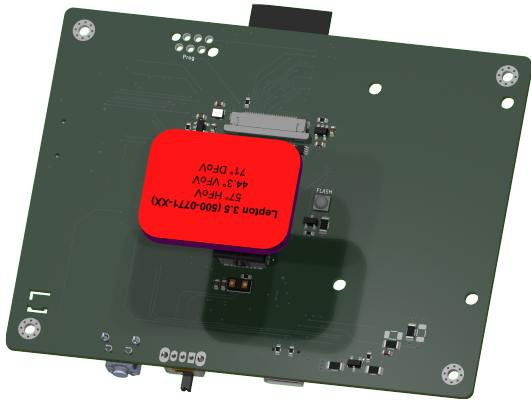
Rev 1

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 23-Feb-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

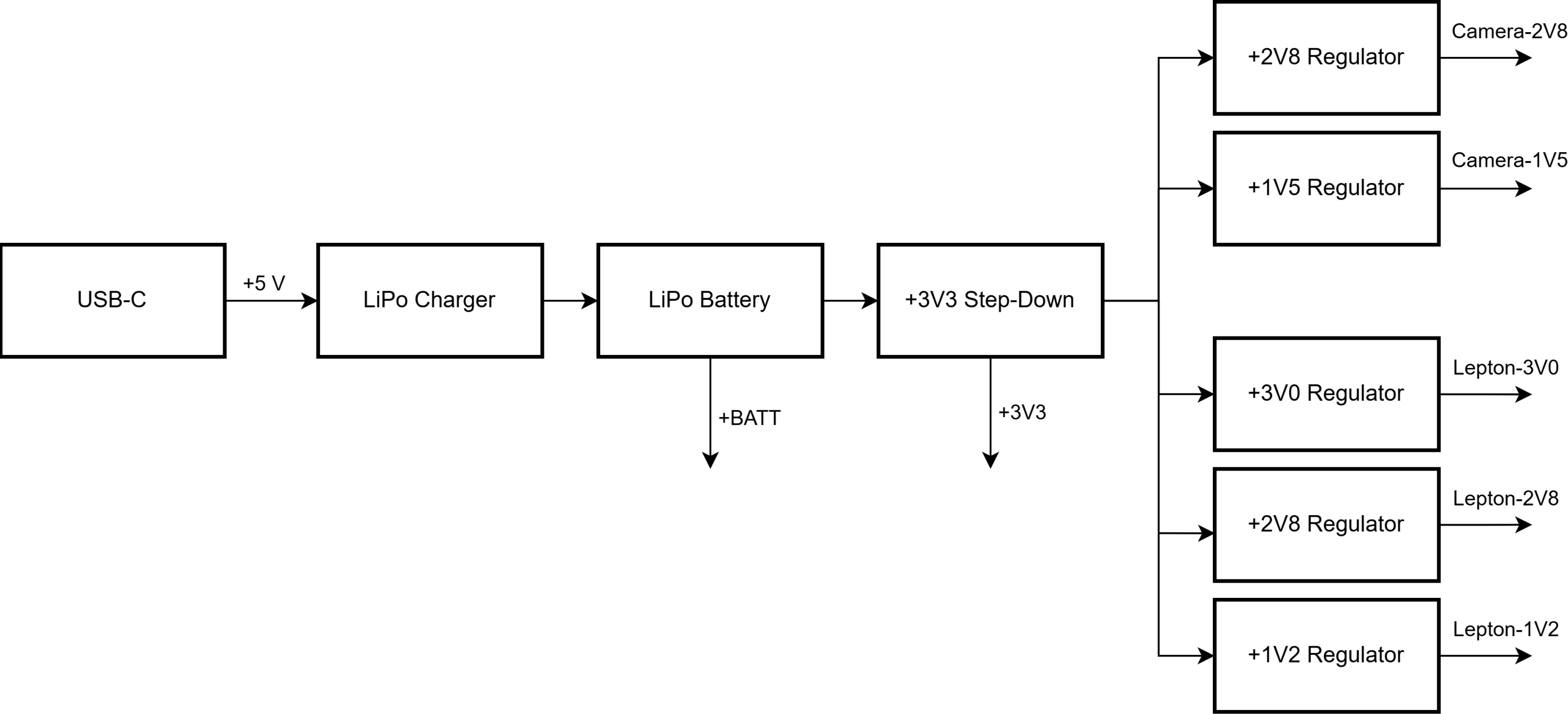
Example text for critical layout guidelines.

	Comments:	Company: PyroVision	Variant: CHECKED	Git Hash: 965ddee
	Board Name: Mainboard	Project Name: PyroVision		
	Sheet Title: Template	File Name: Mainboard.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19
	Sheet Path: /		Reviewer:	Size: A3
				Revision: 1
				Sheet: 1 of 15

1	2	3	4	5	6	7	8
A	[2] Block Diagram						
B							
C							
D							
E							
F							

	Comments:	Company: PyroVision		Variant: CHECKED	Git Hash: 965ddce
		Board Name: Mainboard		Project Name: PyroVision	
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19	Revision: 1
	Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 15

[3] Power - Topology

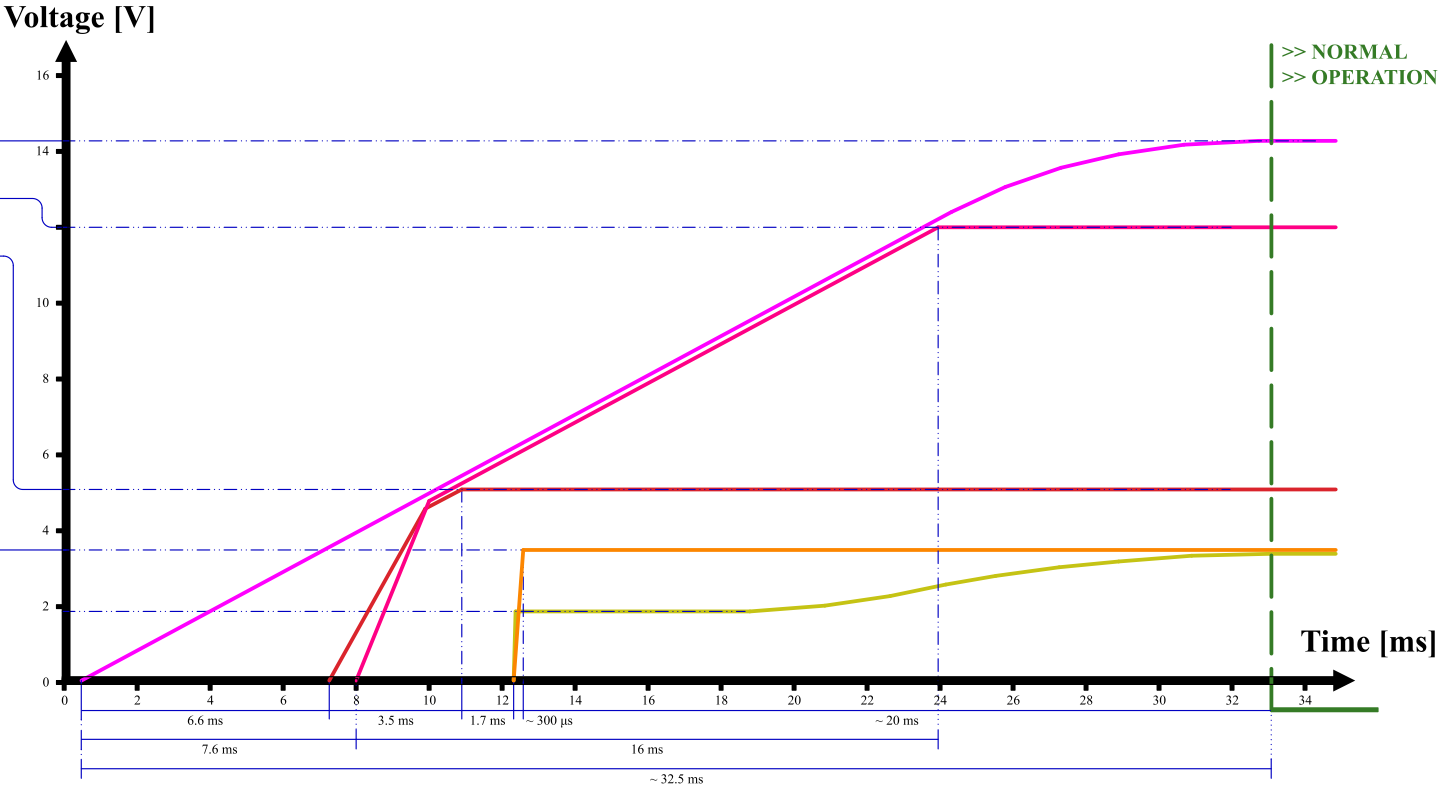


Target specifications:	
Input voltage:	5 V
Max. Input Current	540 mA

	Comments:	Company: PyroVision		Variant: CHECKED	Git Hash: 965ddee
		Board Name: Mainboard		Project Name: PyroVision	
	Sheet Title: Power - Topology	File Name: Power - Topology.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19	Revision: 1
	Sheet Path: /Power - Topology/		Reviewer:	Size: A3	Sheet: 3 of 15

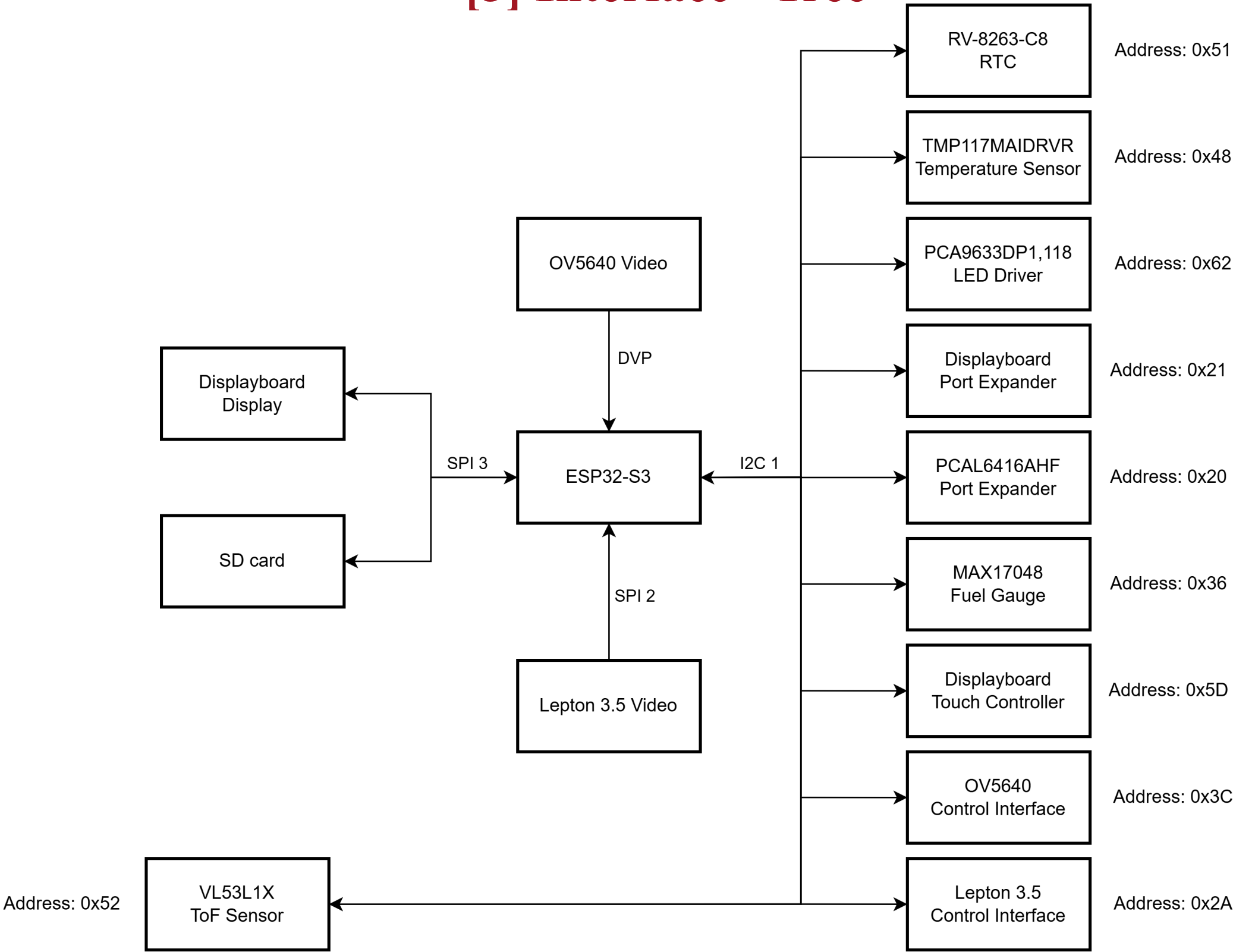
[4] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



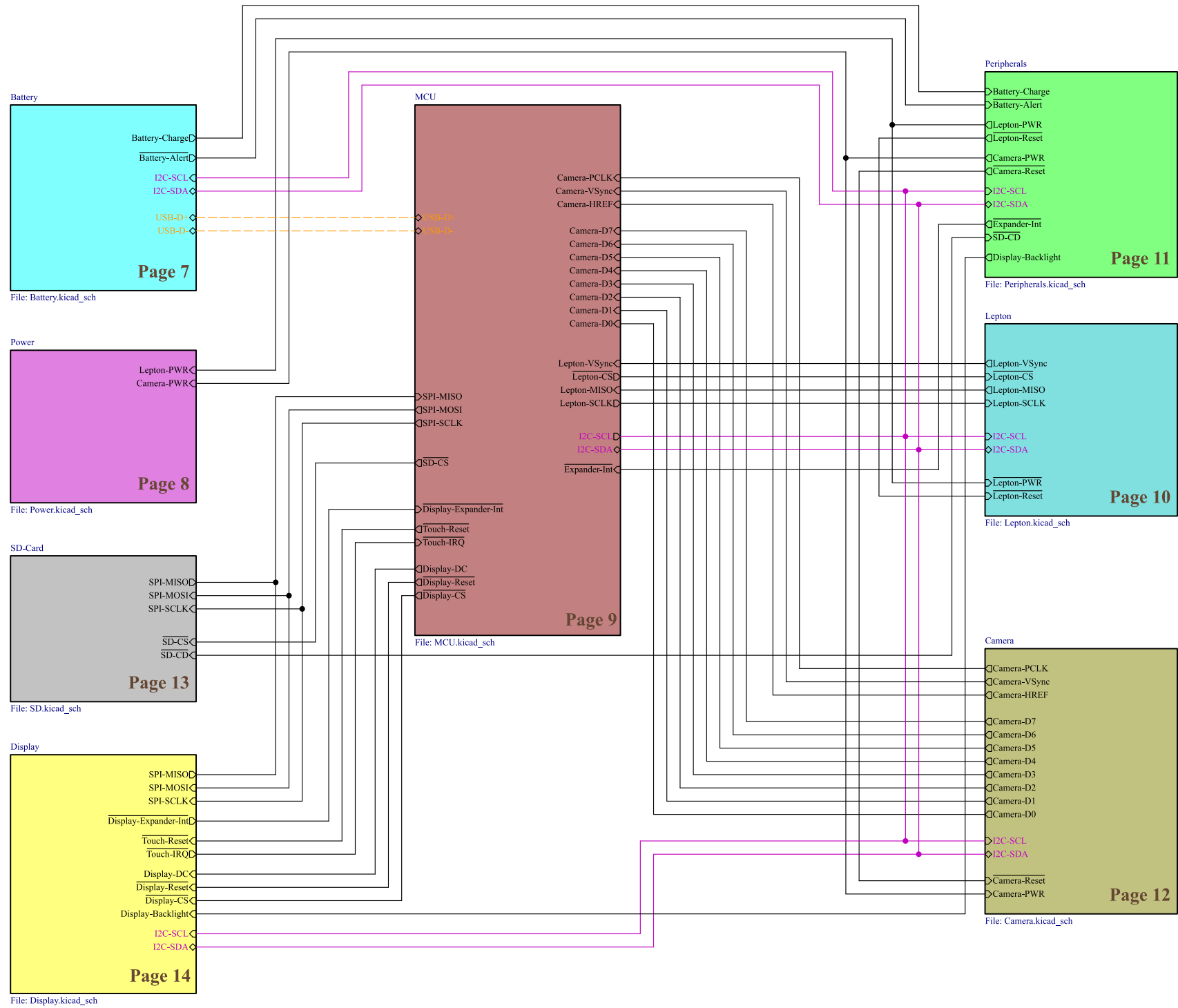
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		Board Name: Mainboard	Project Name: PyroVision	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19
	Sheet Path: /Power - Sequencing/	Reviewer:	Size: A4	Revision: 1
			Sheet: 4 of 15	

[5] Interface - Tree



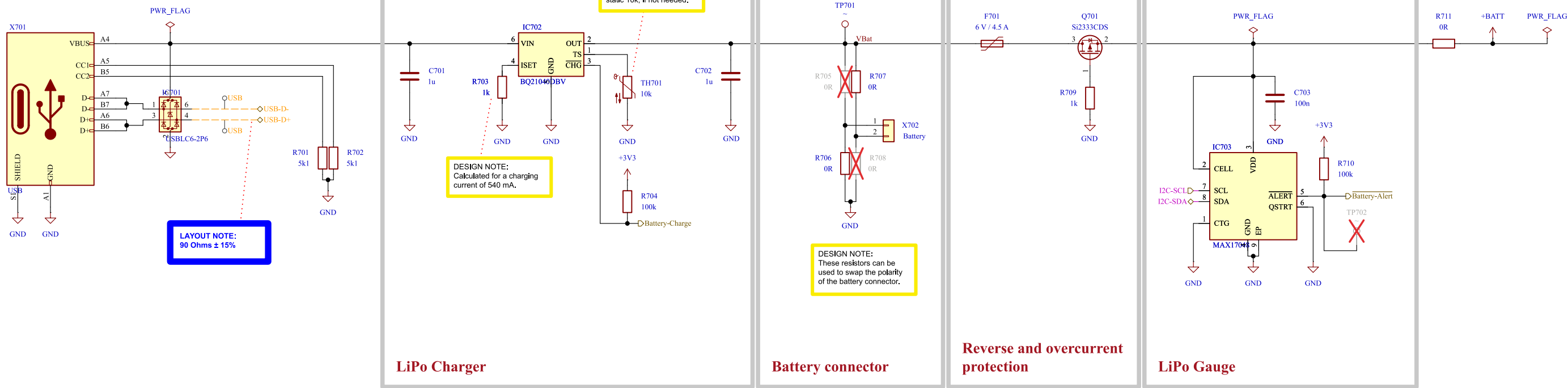
Comments:	Company: PyroVision		Variant: CHECKED	Git Hash: 965ddee
	Board Name: Mainboard		Project Name: PyroVision	
Sheet Title: Interface - Tree	File Name: Interface-Tree.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19	Revision: 1
Sheet Path: /Interface - Tree/		Reviewer:	Size: A3	Sheet: 5 of 15

[6] Project Architecture



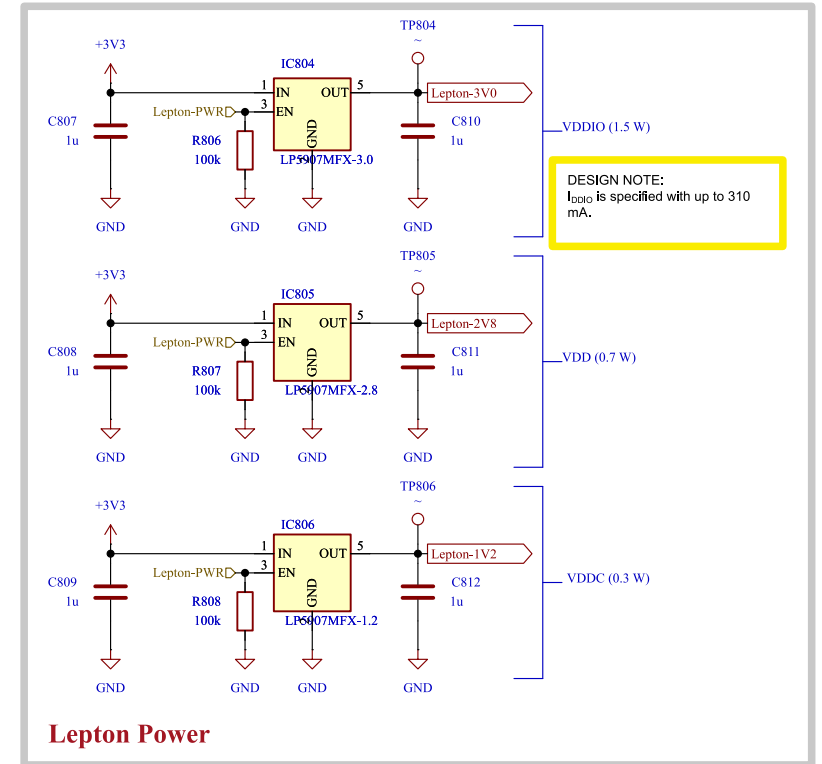
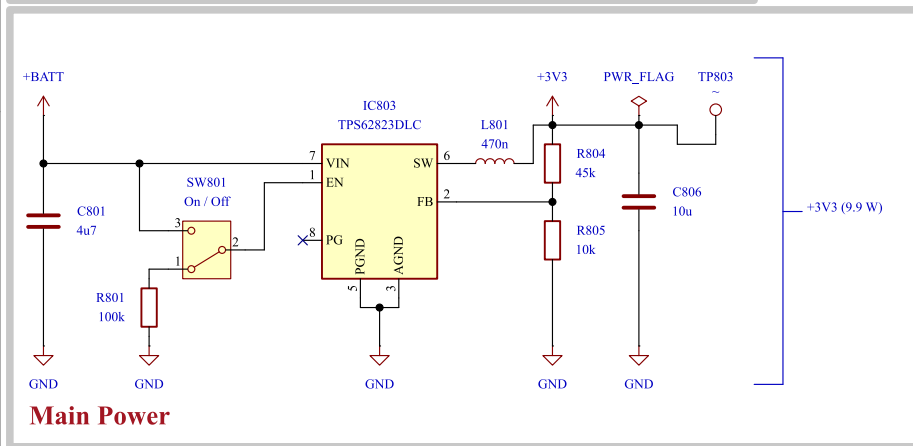
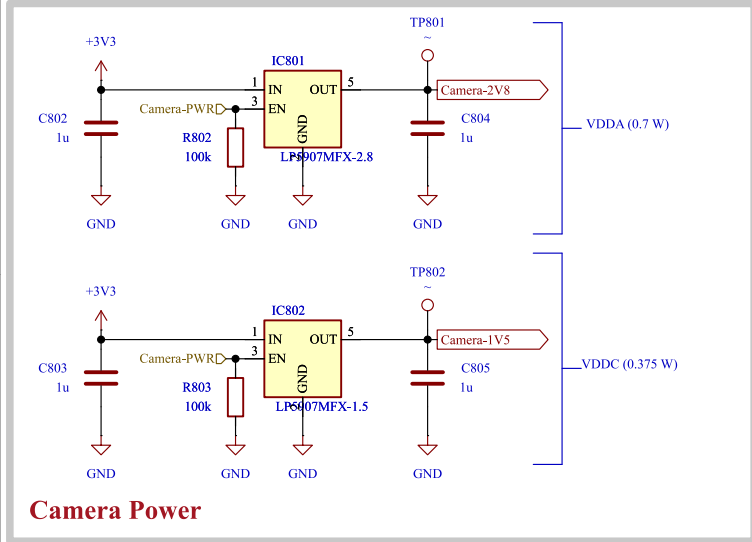
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			Board Name: Mainboard		Project Name: PyroVision	
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19	Revision: 1	
	Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 6 of 15	

[7] Battery



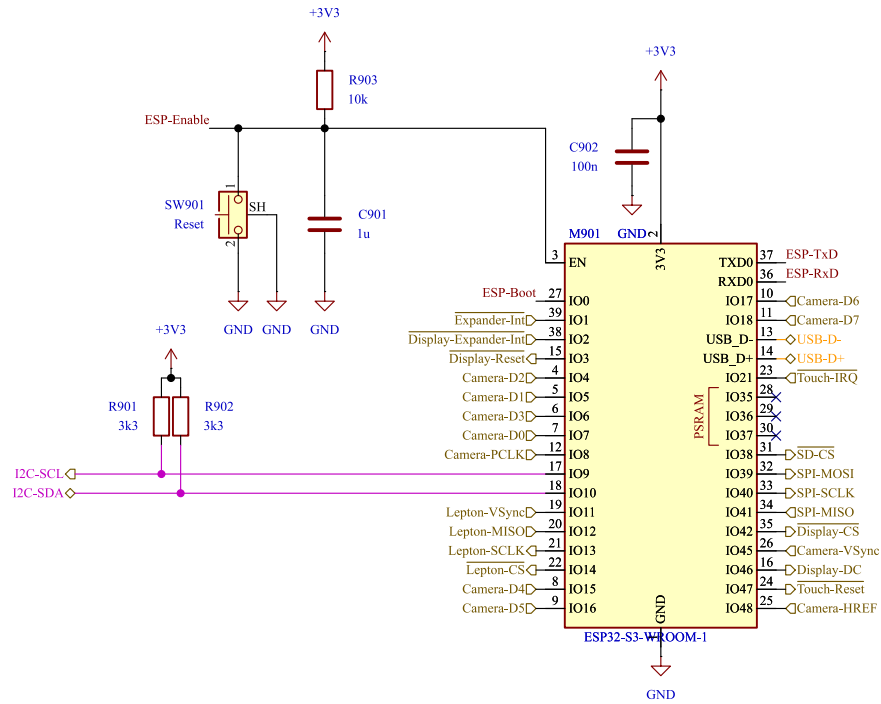
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			Board Name: Mainboard		Project Name: PyroVision	
	Sheet Title: Battery	File Name: Battery.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19	Revision: 1	
	Sheet Path: /Project Architecture/Battery/		Reviewer:	Size: A3	Sheet: 7 of 15	

[8] Power



Comments:	Company: PyroVision		Variant: CHECKED	Git Hash: 965dde
	Board Name: Mainboard		Project Name: PyroVision	
Sheet Title: Power	File Name: Power.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19	Revision: 1
Sheet Path: /Project Architecture/Power/		Reviewer:	Size: A4	Sheet: 8 of 15

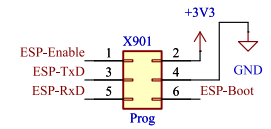
[9] MCU



DESIGN NOTE:
GPIO3 is ignored as a
strapping pin by default.

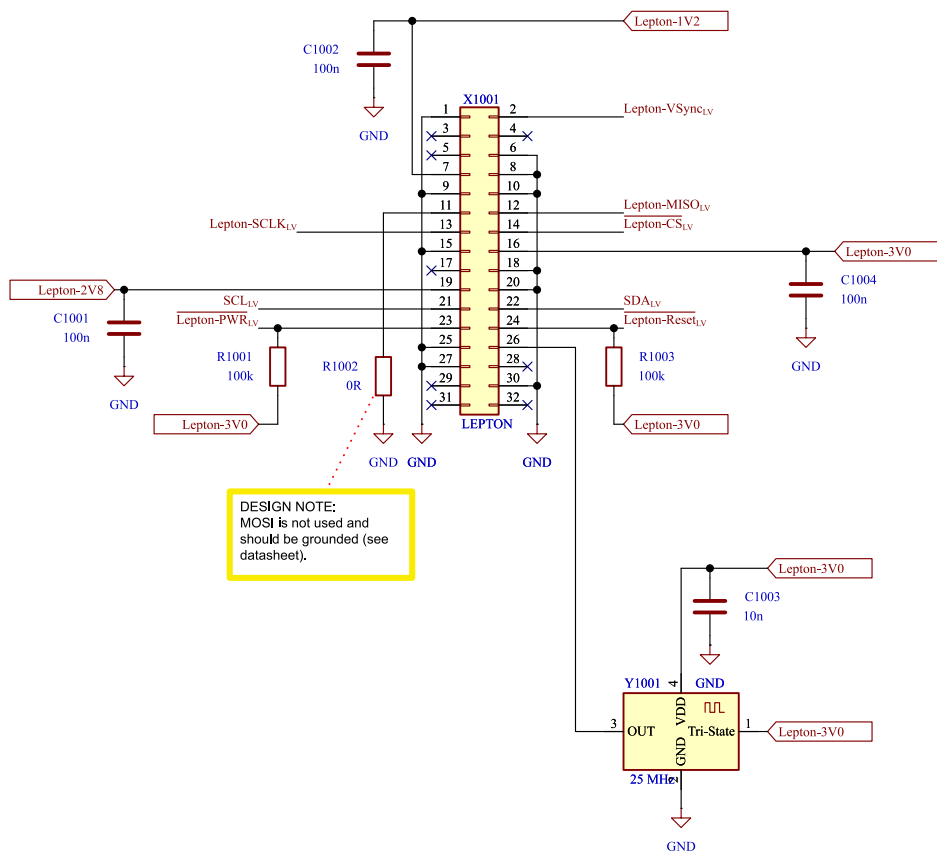
DESIGN NOTE:
GPIO45 must be 0 during boot.

DESIGN NOTE:
GPIO46 must be 0 in Download
Mode (GPIO0 = 0).

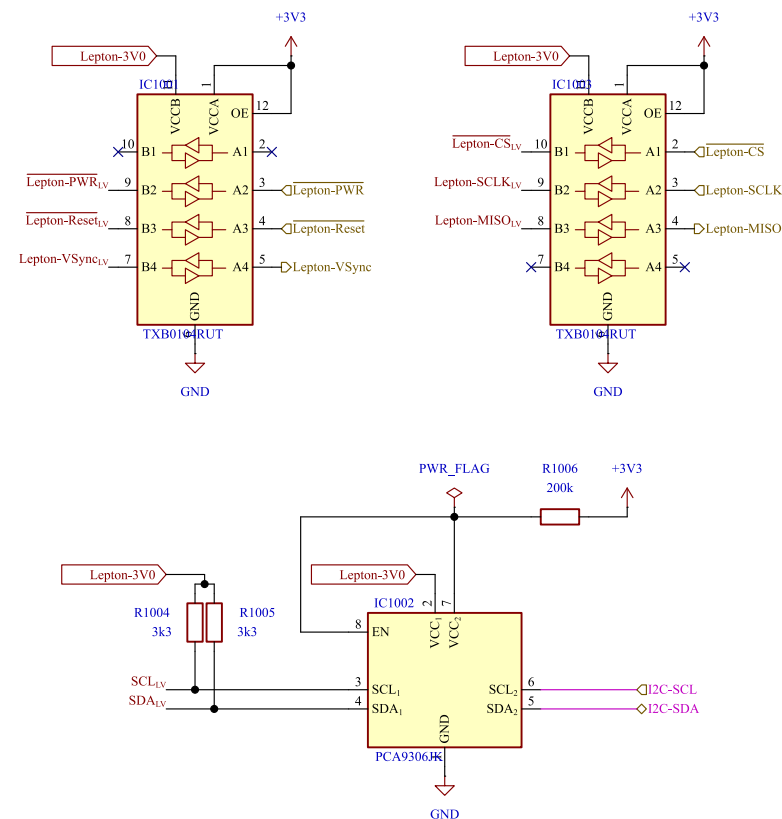


	Comments:	Company: PyroVision		Variant: CHECKED	Git Hash: 965dde
		Board Name: Mainboard		Project Name: PyroVision	
	Sheet Title: MCU	File Name: MCU.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19	Revision: 1
	Sheet Path: /Project Architecture/MCU/		Reviewer:	Size: A4	Sheet: 9 of 15

[10] Lepton



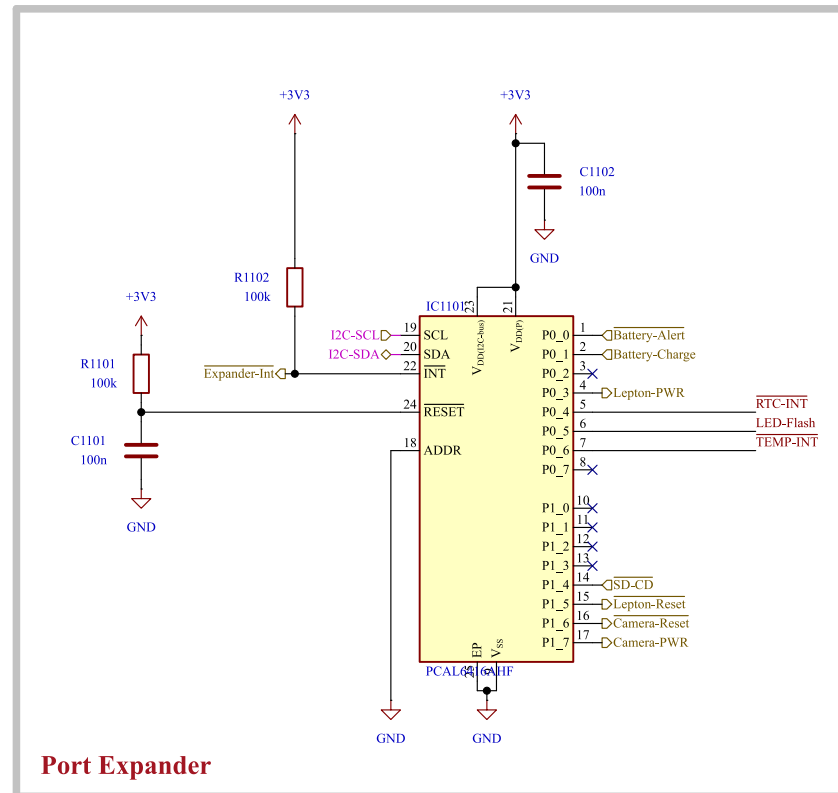
Lepton 3.5 camera connector



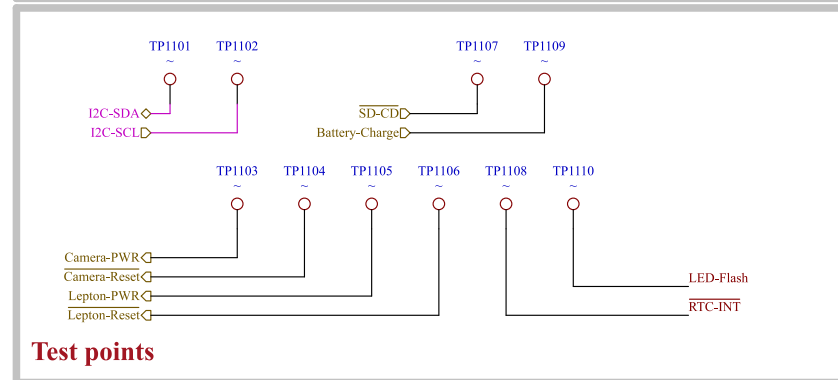
Level shifting (Lepton IO $\leftrightarrow$ System Voltage)

Comments:	Company: PyroVision		Variant: CHECKED	Git Hash: 965dde
	Board Name: Mainboard		Project Name: PyroVision	
Sheet Title: Lepton	File Name: Lepton.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19	Revision: 1
Sheet Path: /Project Architecture/Lepton/		Reviewer:	Size: A4	Sheet: 10 of 15

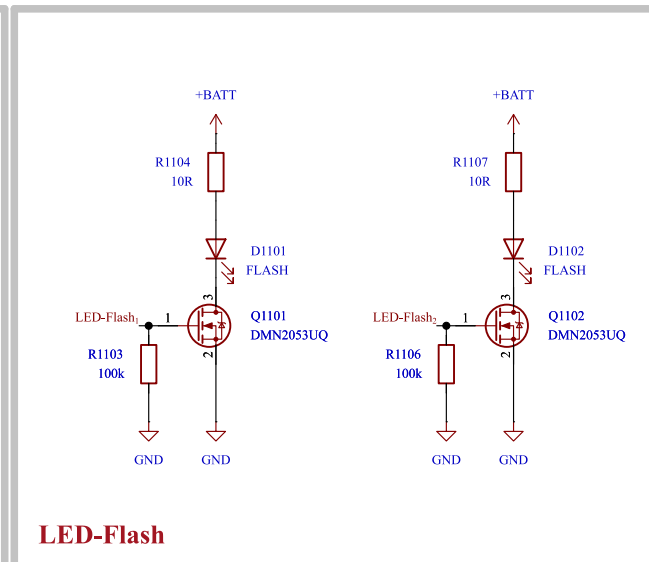
[11] Peripherals



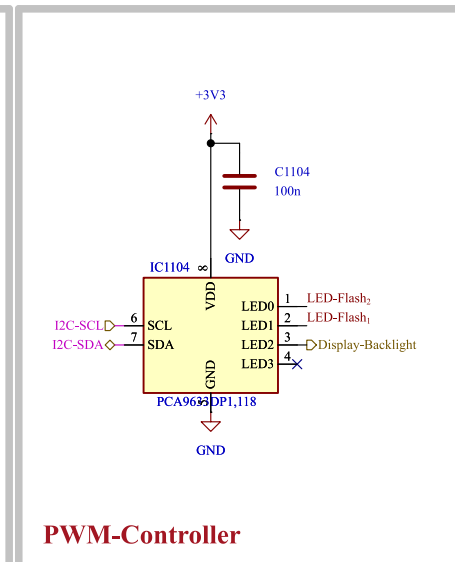
Port Expander



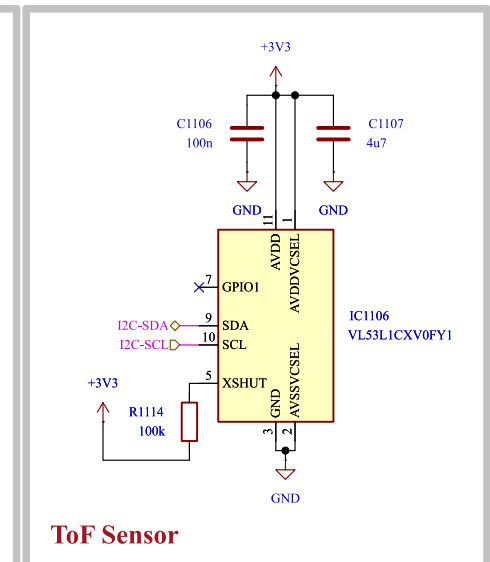
Test points



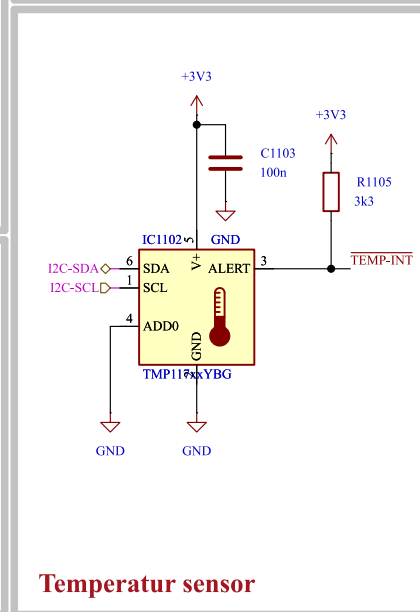
LED-Flash



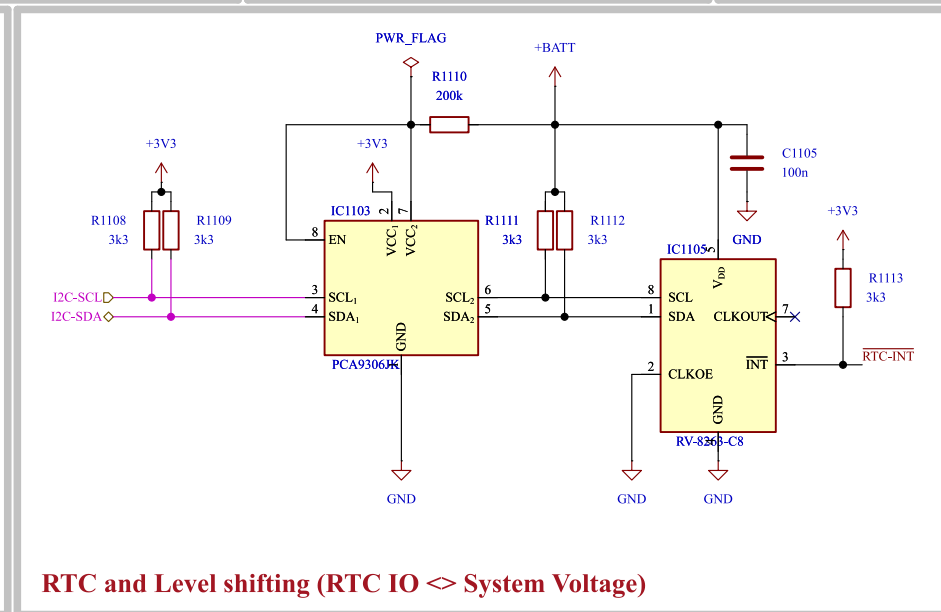
PWM-Controller



ToF Sensor



Temperatur sensor

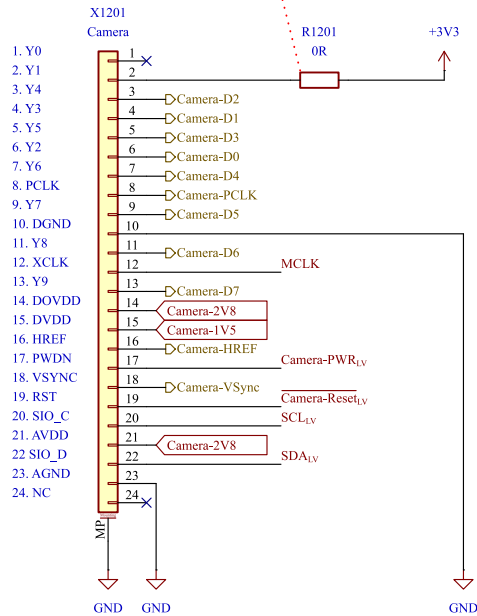


RTC and Level shifting (RTC IO <-> System Voltage)

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			PyroVision		CHECKED	965ddee
			Board Name:		Project Name:	
			Mainboard		PyroVision	
Sheet Title:	Peripherals		File Name:	Peripherals.kicad_sch	Designer:	Date:
					Daniel Kampert	2025-12-19
Sheet Path:	/Project Architecture/Peripherals/		Reviewer:		Size:	Revision:
					A3	11 of 15

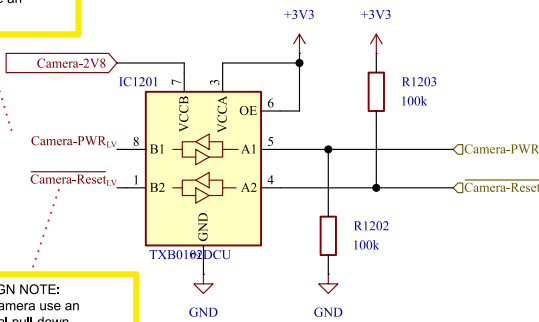
[12] Camera

DESIGN NOTE:
This pin is used for the AF
power supply.



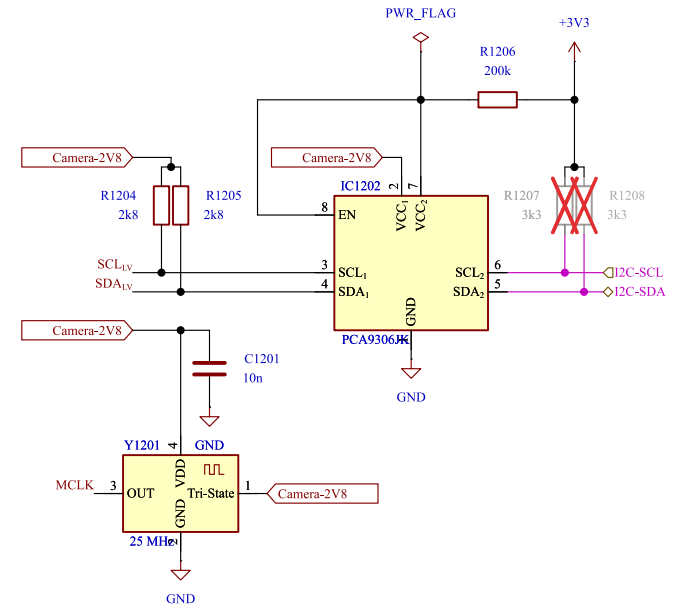
OV5640 camera connector

DESIGN NOTE:
The camera use an
internal pull-up.



DESIGN NOTE:
The camera use an
internal pull-down.

Level shifting (Camera IO <=> System Voltage)



Comments:

Company:

PyroVision

Variant:

CHECKED

Git Hash:

965dde

Board Name:

Mainboard

Project Name:

PyroVision

Sheet Title:

Camera

File Name:

Camera.kicad_sch

Designer:

Daniel Kampert

Date:

2025-12-19

Revision:

1

Sheet Path:

/Project Architecture/Camera/

Reviewer:

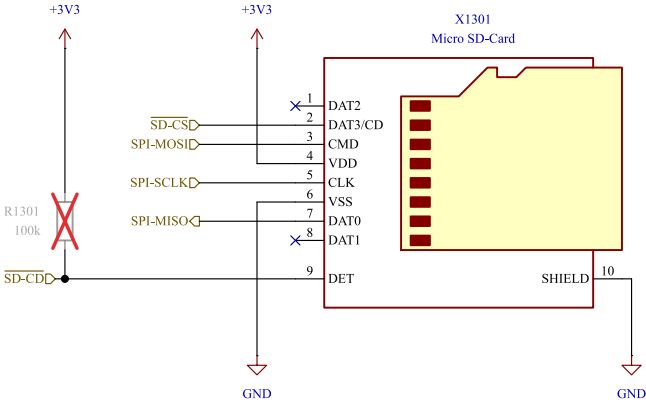
Size:

A4

Sheet:

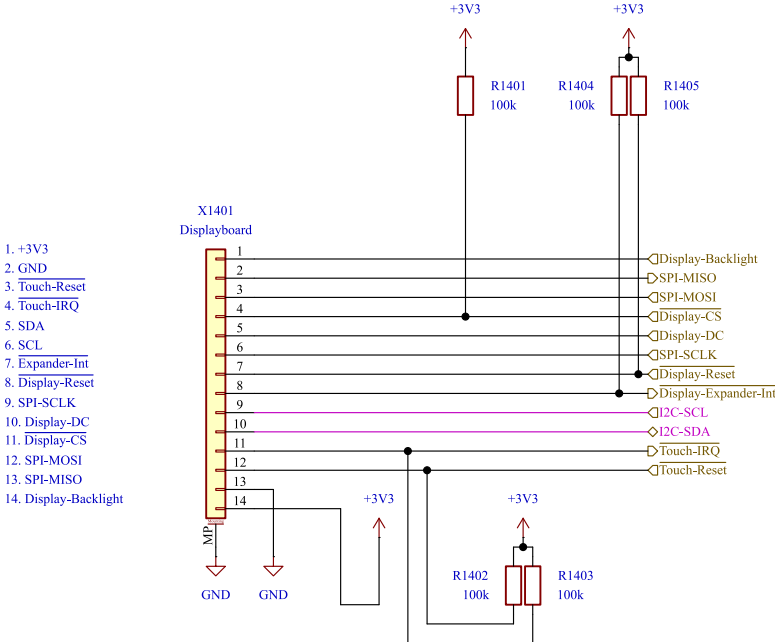
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[13] SD-Card



	Comments:	Company: PyroVision		Variant: CHECKED	Git Hash: 965dde
		Board Name: Mainboard		Project Name: PyroVision	
	Sheet Title: SD-Card	File Name: SD.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19	Revision: 1
	Sheet Path: /Project Architecture/SD-Card/		Reviewer:	Size: A4	Sheet: 13 of 15

[14] Display



	Comments:	Company: PyroVision		Variant: CHECKED	Git Hash: 965ddee
		Board Name: Mainboard		Project Name: PyroVision	
	Sheet Title: Display	File Name: Display.kicad_sch	Designer: Daniel Kampert	Date: 2025-12-19	Revision: 1
	Sheet Path: /Project Architecture/Display/		Reviewer:	Size: A4	Sheet: 14 of 15

[15] Revision History

Release Title	Release Body	Release Title	Release Body
\$(RELEASE_TITLE_1.0.0)	\$(RELEASE_BODY_1.0.0)	\$(RELEASE_TITLE_1.0.1)	\$(RELEASE_BODY_1.0.1)
\$(RELEASE_TITLE_1.0.2)	\$(RELEASE_BODY_1.0.2)	\$(RELEASE_TITLE_1.1.0)	\$(RELEASE_BODY_1.1.0)

Comments:	Company:	Variant:	Git Hash:
	PyroVision	CHECKED	965ddee
	Board Name:	Project Name:	
	Mainboard	PyroVision	
Sheet Title:	File Name:	Designer:	Date:
Revision History	Revision History.kicad_sch	Daniel Kampert	2025-12-19
Sheet Path:	Reviewer:	Size:	Revision:
/Revision History/		A4	1

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